

Pokhara University
Faculty of Science and Technology

Course Code: CMP 645 (3 Credits)
Course Title: Digital Ecosystem (3-0-0)
Nature of the course: Theory
Level: Master

Full marks: 100
Pass marks: 60
Total Lectures: 45 hrs
Program: MCE/ MSc CS

1. Course Description:

This course explores the comprehensive concept of digital ecosystems, covering their evolution, theoretical foundations, architecture, governance models, enabling technologies, sector-specific applications, and emerging trends. Students engage with real-world examples and practical assignments to understand and apply digital ecosystem strategies effectively.

2. General Objectives:

This course aims to equip students with an in-depth understanding of digital ecosystems, their components, management practices, and strategic implications, enabling students to apply ecosystem thinking effectively across business, governmental, and societal contexts.

3. Methods of Instruction:

The course employs interactive lectures, class discussions, group assignments, case studies, and presentations. Students will conduct research on various topics, prepare reports, and engage in practical design and analytical exercises.

4. Contents in Detail

Specific Objectives	Contents
Define digital ecosystems and identify their key characteristics	Unit 1: Introduction to Digital Ecosystems (4 hrs) 1.1 Definition and Key Characteristics 1.2 Evolution from Traditional Ecosystems to Digital Ecosystems 1.3 Components: Platform, Participants, Data, and Value 1.4 Real-world Examples (e.g., Amazon, Google, Alibaba, Smart Cities) 1.5 Overview of Nepal's digital journey (Digital Nepal Framework)
Apply theoretical concepts to analyze digital ecosystems	Unit 2: Theoretical Foundations (5 hrs) Introduction to Ecosystem Theory (Complex Systems Theory, Platform Theory, Network Effects Theory, Co-Evolution Theory, Innovation Ecosystem Theory)
Analyze open versus closed ecosystem architectures	Unit 3: Architecture and Design of Digital Ecosystems (6 hrs) 3.1 Layers: Physical, Application, Service, and Business Layers 3.2 Open vs Closed Ecosystems 3.3 API Economy and Interoperability 3.4 Data Flow and Integration in Ecosystems
Assess governance implications in managing digital ecosystems	Unit 4: Governance and Management (5 hrs) 4.1 Ecosystem Governance Models (Centralized Governance Model, Decentralized Governance

	Model, Hybrid Governance Model) 4.2 Platform Leadership and Orchestration 4.3 Trust, Security, and Privacy Issues
Evaluate how emerging technologies support ecosystem functionalities	Unit 5: Technologies Enabling Digital Ecosystems (6 hrs) 5.1 Internet of Things (IoT) and Edge Computing 5.2 Artificial Intelligence and Machine Learning (Application of AI in Digital Ecosystem) 5.3 Blockchain and Decentralization 5.4 Cloud Computing and Microservices
Identify strategic advantages of digital ecosystems in various sectors	Unit 6: Digital Ecosystems in Different Sectors (6 hrs) 6.1 E-commerce and Retail Ecosystems 6.2 Health and Smart City Ecosystems 6.3 Education Ecosystems (MOOCs, EdTech) 6.4 Tourism and Hospitality Ecosystems 6.5 Industrial and Manufacturing Ecosystems (Industry 4.0)
Apply ecosystem mapping and strategic positioning tools	Unit 7: Business Models and Strategy (5 hrs) 7.1 Platform Business Models (Transaction Platform Model, Advertising Platform Model, Data-Driven Platform Model, Product-as-a-Service) Platform Model, Social Platform Model) 7.2 Ecosystem Mapping and Strategic Positioning 7.3 Metrics for Ecosystem Health and Success 7.4 Case Studies of Failures and Successes (e.g., Nokia, Uber, Tesla, Panasonic, KODAK)
Evaluate the impact of emerging trends on future digital ecosystems	Unit 8: Emerging Trends and Future Directions (4 hrs) 8.1 Metaverse and Immersive Ecosystems 8.2 Ecosystemization of AI 8.3 Sustainable Digital Ecosystems (Green IT) 8.4 Digital Twin Ecosystems 8.5 Research Frontiers and Open Challenges

Practical Component / Assignments:

- **Case Study Analysis:** Choose a real-world domain digital ecosystem (e.g.: health, education, governance, fintech)
- **Ecosystem Design Project:** Design a digital ecosystem for a sector (e.g., agriculture, tourism, finance)
- **Research Paper / White Paper:** Investigate a current trend (e.g., DAOs in ecosystems, platform monopolies)
- **Group Presentation:** Emerging Technologies Shaping Future Ecosystems

5. Evaluation system and Students' Responsibilities

Internal Evaluation

The internal evaluation of a student may consist of assignments, attendance, internal assessment and group workshop. The internal evaluation scheme for this course is as follows:

Internal Evaluation	Weight	Marks	External Evaluation	Marks
Theory			Semester-End examination	40
Attendance & Class Participation	10%	6		
Assignments	10%	6		
Presentations/Quizzes	20%	12		
Internal Assessment	40%	24		
Group Workshops	20%	12		
Total Internal		60		
Full Marks: 60 + 40 = 100				

Student Responsibilities:

Each student must secure at least 60% marks separately in internal assessment and practical evaluation with 80% attendance in the class in order to appear in the Semester End Examination. Failing to get such a score will be given NOT QUALIFIED (NQ) to appear for the Semester-End Examinations. Students are advised to attend all the classes, formal exam, test, etc. and complete all the assignments within the specified time period. Students are required to complete all the requirements defined for the completion of the course.

6. Prescribed Books and References

Textbooks:

- Parker, G. G., Van Alstyne, M. W., & Choudary, S. P. (2016). Platform revolution: How networked markets are transforming the economy and how to make them work for you. W. W. Norton & Company.
- Cusumano, M. A., Gawer, A., & Yoffie, D. B. (2019). The business of platforms: Strategy in the age of digital competition, innovation, and power. Harper Business.
- Corallo, A., Passiante, G., & Prencipe, A. (2007). Digital business ecosystems. Edward Elgar Publishing.

Reference Books:

- Tiwana, A. (2013). Platform ecosystems: Aligning architecture, governance, and strategy. Morgan Kaufmann.
- Adner, R. (2012). The wide lens: What successful innovators see that others miss. Portfolio Penguin.
- Easley, D., & Kleinberg, J. (2010). Networks, crowds, and markets: Reasoning about a highly connected world. Cambridge University Press.